GBOH-IGBARA D. CHARLES

AI Researcher & Developer | Data Scientist | MSc IT Applicant

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PROFESSIONAL PROFILE

Results-driven AI Researcher, Data Scientist, and Health Technology Developer with 4+ years of experience spanning machine learning, statistical modelling, NLP, and clinical AI system deployment. Dual published researcher with a peer-reviewed paper in the Open Access Research Journal of Multidisciplinary Studies and a preprint on explainable AI in cardiovascular healthcare submitted to medRxiv. Author and sole developer of CardioAI — a live, deployed explainable AI system for cardiovascular risk prediction and patient retention prediction, accessible at cardioai-joihealth.streamlit.app. Recognized with the 3MTT DeepTech Showcase Award for innovative ML capstone work. Seeking admission to an MSc in Information Technology programme to formalise and extend expertise in clinical AI, health informatics, and intelligent system design.

KEY ACHIEVEMENTS & AWARDS

- 3MTT DeepTech Showcase Award — Recognized for innovative ML capstone project among national cohort (2025)
- DeepTech DSN Fellowship — Fully funded Data Science & Machine Learning fellowship (2025–2026)
- Published Researcher — Peer-reviewed paper in Open Access Research Journal of Multidisciplinary Studies (2024)
- medRxiv Preprint Author — Cardiovascular AI paper indexed on global preprint server, reference MEDRXIV/2026/349630
- Live Deployed AI System — CardioAI available at cardioai-joihealth.streamlit.app — publicly accessible worldwide
- CareerEx & Access Bank Youthrive — Data Analysis & Visualization recognition (2025)

RESEARCH & PUBLICATIONS

Publication 1 — Peer-Reviewed Journal Article

Gboh-Igbara D. Charles, Bassey Edet Effiong, Sebastian Stephen Akpan, Ugbe Thomas Adidaumbe, Enang Ekaette Inyang, Christian Elendu Onwuke, Joy Uket Ogum, Nja Egom Mbe. *(2024). Application of Logistic Regression Model on the Spread of Malaria Infection in Calabar Municipality: A Case Study of University of Calabar Teaching Hospital.* Open Access Research Journal of Multidisciplinary Studies, 07(01), 022–038. DOI: https://doi.org/10.53022/oarjms.2024.7.1.0055

Received: 20 November 2023 | Revised: 23 January 2024 | Accepted: 26 January 2024 | Department of Mathematics and Statistics, Faculty of Physical Sciences, University of Calabar

- Applied logistic regression statistical modelling to predict and analyse malaria infection spread patterns in a Nigerian teaching hospital using real clinical data
- Conducted full statistical analysis including odds ratios, confidence intervals, and model diagnostics on a multi-author collaborative study
- Contributed to evidence-based public health research bridging statistical methodology and clinical practice in a Nigerian healthcare context

Publication 2 — medRxiv Preprint (Under Journal Review)

Gboh-Igbara D. Charles. (2025). *CardioAI: An Explainable Machine Learning System for Cardiovascular Risk Prediction and Patient Retention in Nigerian Healthcare Settings.* medRxiv Preprint Server. Reference: MEDRXIV/2026/349630. Live System: cardioai-joihealth.streamlit.app | GitHub: github.com/gbohigbaradc/cardioai-project

- Sole developer and researcher of CardioAI — a full end-to-end explainable AI clinical decision support system deployed live on the internet
- Trained 4 cardiovascular risk classifiers (Logistic Regression, Random Forest, XGBoost, Neural Network) achieving AUC-ROC of 0.999 on held-out test set; cross-validated AUC of 0.97 confirming generalisation
- Engineered the novel Lifestyle Risk Index — an original composite modifiable risk score combining 5 clinical markers, validated as the 3rd most important SHAP feature globally
- Implemented SHAP explainability providing per-patient feature attribution readable by non-specialist clinicians — directly addressing the black-box adoption barrier in clinical AI
- Built a patient retention prediction module identifying operational dropout risk factors and generating specific clinical interventions for each at-risk patient
- Developed a full NLP and OCR pipeline using Tesseract v5.5 and spaCy to extract structured data from handwritten clinical notes, scanned images, and PDFs
- Deployed as a publicly accessible interactive Streamlit web application — currently live and accessible worldwide
- Submitted to medRxiv with 28 academic references; targeting PLOS ONE and BMC Medical Informatics for full journal submission

PROFESSIONAL EXPERIENCE

Information Officer — AI & Data Systems | **JoiHealth, Lagos** *Jan 2026 – Present*

- Led IT infrastructure maintenance and Hospital Management System (HMS) administration for a luxury medical facility
- Applied machine learning and data analytics to optimise procurement workflows and financial reporting, reducing reporting cycle time by 30%
- Managed RDBMS systems ensuring integrity, security, and compliance of all medical records and clinical workflows
- Developed and deployed CardioAI — a live clinical AI system for cardiovascular risk assessment — as part of the organisation's digital health strategy

Data Science & ML Research Fellow | **3MTT | DSN DeepTech Fellowship** *Sep 2025 – Feb 2026*

- Built predictive models using Python (scikit-learn, TensorFlow) for safety risk and operational analysis in a structured research fellowship
- Automated analytical reporting workflows using generative AI tools including Microsoft Copilot and GPT-based pipelines
- Collaborated across distributed teams using Azure ML, GitHub, and Trello in a fully remote research environment
- Received the 3MTT DeepTech Showcase Award for outstanding ML capstone project among national cohort

Field Operations Analyst | **Greenville LNG, Port Harcourt** *Nov 2024 – Nov 2025*

- Designed and maintained Power BI dashboards for real-time safety metrics and operational performance monitoring
- Digitised inspection records and equipment datasets, improving data accessibility and retrieval efficiency by 40%
- Annotated industrial equipment datasets for predictive maintenance machine learning models

Data Analytics & ML Instructor | **Emblic Technologies Ltd** *Sep 2023 – Jan 2024*

- Designed and delivered training curriculum in Python, SQL, Excel, and data visualisation for junior analysts
- Led practical workshops on data cleaning, feature engineering, statistical analysis, and predictive modelling

Data Analyst Trainee | **NJFP (3MTT Cohort 3)** *May 2023 – May 2024*

- Developed dashboards and management reports using Power BI, Excel, and SQL for KPI tracking and executive decision-making

- Supported data storytelling initiatives translating complex analytical outputs into actionable business recommendations

EDUCATION

B.Sc. (Hons) Statistics — Second Class Upper | **University of Calabar, Nigeria** *Graduated Feb 2021*

- Department of Mathematics and Statistics, Faculty of Physical Sciences
- Final year research on statistical modelling of infectious disease spread — foundation for published malaria logistic regression study
- Relevant coursework: Probability & Statistical Inference, Regression Analysis, Multivariate Methods, Biostatistics, Operations Research

NYSC Discharge Certificate | **National Youth Service Corps, Nigeria** *Feb 2022*

- Successfully completed national service — demonstrated civic responsibility and professional commitment

TECHNICAL SKILLS

AI & Machine Learning	scikit-learn, XGBoost, TensorFlow, SHAP, SMOTE, Neural Networks (MLP), GridSearchCV, cross-validation
NLP & OCR	spaCy, NLTK, Tesseract OCR v5.5, Regex NLP pipelines, pdfplumber, clinical entity extraction
Data Analytics	Python (Pandas, NumPy, Matplotlib, Seaborn), SQL, Power BI, Tableau, Microsoft Excel
Deployment & MLOps	Streamlit, Streamlit Community Cloud, GitHub, Git version control, REST API design, joblib model serialisation
Cloud & Databases	Azure ML, GCP BigQuery, PostgreSQL, MySQL, SQL Server, RDBMS administration
Programming	Python (advanced), SQL (intermediate), JavaScript (basic), Bash scripting
Research Methods	Logistic regression, SHAP explainability, feature engineering, synthetic data generation, AUC-ROC evaluation, Brier Score
Tools & Collaboration	GitHub, Jupyter Notebook, VS Code, Anaconda, Slack, Trello, Zoom, Microsoft Teams

CERTIFICATIONS & TRAINING

- Data Science, Machine Learning, NLP, Computer Vision, Cloud Computing — 3MTT/DSN DeepTech Fellowship (2025)
- Digital For All Initiative (DFA 2.0) — Data Science & Machine Learning Foundations (2025)
- CareerEx & Access Bank Youthrive — Data Analysis & Visualization (2025)
- NLNG/NCDMB HCCD Train 7 Program — National energy sector capacity development

KEY PROJECTS

CardioAI — Explainable AI Clinical Decision Support System

Live: cardioai-joihealth.streamlit.app | Code: github.com/gbohigbaradc/cardioai-project | Paper: MEDRXIV/2026/349630

- Sole developer of a full end-to-end clinical AI system combining cardiovascular risk prediction, patient retention prediction, SHAP explainability, and clinical NLP/OCR processing

- Stack: Python, XGBoost, Random Forest, SHAP, spaCy, Tesseract OCR, Streamlit, GitHub — deployed live on Streamlit Community Cloud
- Best model AUC-ROC: 0.999 (XGBoost) | Cross-validated AUC: 0.97 | 1,025 training records | 12 output charts | 5-page interactive dashboard

Malaria Spread Statistical Modelling — University of Calabar Teaching Hospital

- Applied logistic regression to model malaria infection spread using real patient data from UCTH — contributed to peer-reviewed publication
- Multi-author academic collaboration within the Department of Mathematics and Statistics, University of Calabar

RESEARCH INTERESTS (MSC FOCUS)

- Explainable AI and clinical decision support systems for low-resource healthcare environments
- Machine learning applications in cardiovascular disease prediction and patient outcome optimisation
- Natural language processing and OCR for unstructured clinical document digitisation in African healthcare settings
- Patient retention modelling and operational analytics in rehabilitation and preventive care
- Health informatics, digital health infrastructure, and AI ethics in medical AI deployment

REFERENCES

Available on request. Academic and professional references from University of Calabar (Statistics Department) and 3MTT/DSN Fellowship supervisors.